Causal Effect of Payment Protection Plans on Credit Default Risk

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April 05, 2026

Abstract

Understanding whether Payment Protection Plans (PPPs) genuinely reduce credit default risk is critical for financial institutions. However, evaluating observational data is challenging because the treatment is not randomized. In this study, we estimate the Average Causal Effect (ACE) of PPP enrollment on credit default. To rigorously control for confounding, we compare a naive full-covariate baseline against a Post-Double-Selection LASSO and two structural causal discovery methods (Rank PC and Fast-IAMB). The ACE is estimated using Outcome Regression (G-computation) and Inverse Probability Weighting (IPW) via logistic regression, with uncertainty quantified through a non-parametric bootstrap. Our findings indicate a statistically significant positive causal effect of PPPs on default risk, consistent with a mechanism of "moral hazard": protection from immediate financial penalties may inadvertently encourage riskier repayment behavior. Furthermore, structurally derived adjustment sets yield superior statistical efficiency compared to high-dimensional predictive models, avoiding practical positivity violations that render high-dimensional IPW estimates uninformative.

1 Introduction and Causal Framework

Banks routinely track whether customers will miss their credit card payments, as defaults on repayments constitute the primary source of interest income for card issuers. To mitigate the risk of default during financial hardship, certain customers may enroll in Payment Protection Plans (PPPs), which allow them to pause repayments without incurring immediate interest charges.

Understanding whether PPPs genuinely reduce default risk is of direct practical importance, yet this question cannot be answered by a naive comparison of enrollees and non-enrollees. Customers who elect to enroll in a PPP are likely to be at a higher *a priori* risk of missing payments, introducing a selection bias. Because of this structural confounding, the true causal effect is distinct from the simple observational difference:

$$\text{Causal Effect} \neq \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0]$$

To formally define the causal estimand, we rely on the Potential Outcomes framework [4]. Let the binary treatment $A \in \{0, 1\}$ denote PPP enrollment, and the binary outcome $Y \in \{0, 1\}$ denote credit default. We define $Y^{a=1}$ and $Y^{a=0}$ as the potential outcomes if a customer were compulsorily enrolled or not enrolled, respectively. The Average Causal Effect (ACE) is defined as:

$$\text{ACE} = \mathbb{E}[Y^{a=1}] - \mathbb{E}[Y^{a=0}]$$

To identify the ACE from observational data, we rely on three core causal assumptions [4]:

1. **Consistency:** The observed outcome matches the potential outcome for the treatment actually received: $Y = AY^{a=1} + (1 - A)Y^{a=0}$.

2. **Conditional Exchangeability:** $Y^a \perp\!\!\!\perp A \mid \mathbf{L}$. Conditional on a set of observed covariates $\mathbf{L}$, the treatment assignment is independent of the potential outcomes. To satisfy this, $\mathbf{L}$ must block all *backdoor paths* (spurious associations driven by common causes) without opening *collider paths*. A collider C is a common effect of two independent variables ($A \to C \leftarrow Y$); conditioning on C induces a spurious dependence between A and Y.

3. **Positivity:** $0 < \mathbb{P}(A = 1 \mid \mathbf{L} = \ell) < 1$ for all ℓ. Every individual must have a strictly positive probability of receiving either treatment. Adjusting for high-dimensional sets $\mathbf{L}$ often violates this assumption in practice (practical positivity violation), leading to extreme estimation variance [4].

This report addresses two questions: (1) does PPP enrollment causally increase or decrease the probability of credit default? (2) how does the choice of adjustment set, informed by causal discovery, affect both the magnitude and precision of the causal estimate? To answer them, we compare four adjustment strategies of increasing theoretical sophistication, a full covariate model, LASSO-based selection, and two causal discovery methods, and evaluate each under two complementary estimators (OR and IPW) whose agreement serves as a robustness check against model misspecification.

2 Dataset Overview

The dataset mimics a real credit repayment study ($n = 500$). The variables are summarized in Table 1. To ensure clarity in the subsequent causal graphs, we specify the financial nomenclature:

- `LIMIT_BAL`: total credit granted (individual and family combined).
- `PAY_0` to `PAY_6`: repayment status for the previous six months on an ordinal scale (-2 = no consumption, -1 = paid in full, 0 = minimum payment, 1 to 9 = months of delay).
- `BILL_AMT1` to `BILL_AMT6`: statement balance billed each month.
- `PAY_AMT1` to `PAY_AMT6`: actual amount paid each month.

Table 1: Summary of dataset variables ($n = 500$).

Variable	**Type**	**Summary / Range**
Missed payment (Y)	Binary	24% defaulted
PPP enrollment (A)	Binary	44% enrolled
Credit limit (`LIMIT_BAL`)	Continuous	[10 000; 650 000]
Age (`AGE`)	Continuous	[21; 73]
Payment status (`PAY_0--6`)	Ordered categ.	-2 to 9 (integer)
Bill amount (`BILL_AMT1--6`)	Continuous	[$-$2,650; 630,458]
Amount paid (`PAY_AMT1--6`)	Continuous	[0; 193,358]
Education	Categorical	Grad., Univ., High School

Preprocessing. The ordinal `PAY` variables are encoded as integers preserving their natural order (higher values indicate longer payment delay). The categorical `EDUCATION` variable is dummy-encoded into three binary indicators (Graduate school as the reference category) for use in the regression models. Two versions of the dataset are maintained throughout: one with the original ordinal encoding for causal discovery algorithms (which require numeric or factor inputs), and one with dummy-encoded education for the logistic regression models used in estimation.

3 Causal Discovery and Adjustment Strategies

Identifying the correct adjustment set $\mathbf{L}$ is the central methodological challenge: conditioning on too few variables leaves residual confounding, while conditioning on too many risks positivity violations and efficiency loss. We evaluate four strategies of increasing theoretical sophistication. Structural methods (Rank PC, Fast-IAMB) recover the Markov blanket of Y from the data-generating mechanism directly, providing a principled answer to which variables must be adjusted for. The data-adaptive LASSO provides a flexible benchmark that does not require distributional assumptions but lacks the graph-theoretic guarantees of structural methods. The Full model serves as a naive upper bound. All analyses use a fixed random seed for reproducibility.

Definition 3.1 (Markov blanket)**.** Let V be the set of all variables in a domain. A Markov Blanket of a target variable $T \in V$, denoted $B(T) \subseteq V \setminus \{T\}$, is any subset of variables such that T is conditionally independent of all remaining variables given $B(T)$ [5]:

$$T \perp\!\!\!\perp (V \setminus (B(T) \cup \{T\})) \mid B(T)$$

A minimal Markov Blanket is called a *Markov Boundary*, corresponding uniquely to the parents, children, and spouses of T in a DAG. Adjusting for the Markov Boundary of Y (excluding the treatment A) is sufficient to block all backdoor paths without over-adjusting.

3.1 Causal Structure via Rank PC Algorithm

Standard implementations of the PC algorithm test conditional independence via partial correlations, which assume the data follows a multivariate normal (Gaussian) distribution. Financial data heavily violates this assumption due to extreme outliers (heavy tails) and discrete categories. To overcome this, we utilize the Rank PC algorithm [3].

Instead of analyzing raw data values, the Rank PC algorithm computes an empirical Spearman rank correlation matrix ($\hat{\mathbf{R}}$). Unlike standard Pearson correlation, which measures the linear relationship between exact numerical values, Spearman's correlation evaluates the monotonic relationship between the relative rankings of the data points (e.g., 1st, 2nd, 3rd largest). This rank-based approach makes it highly robust to extreme financial outliers. We then apply a mathematical transformation to this matrix:

$$\hat{\Sigma}_{ij} = 2 \sin\left(\frac{\pi}{6}\,\hat{R}_{ij}\right)$$

This transformation acts as a Gaussian copula, effectively mapping the rank correlations into a latent Gaussian space. This allows the standard PC conditional independence tests to remain valid without requiring the underlying data to be normally distributed.

The algorithm then iteratively removes and orients edges (using a significance level of $\alpha = 0.05$), yielding a Completed Partially Directed Acyclic Graph (CPDAG). Finally, our adjustment set is defined as the Markov Boundary of the outcome Y on this graph (its parents, children, and spouses), explicitly excluding the treatment variable A.

3.2 Markov Blanket Discovery via Fast-IAMB

Rather than learning the full graph, Fast-IAMB [7] directly targets the Markov Boundary of Y, which is sufficient for adjustment without the computational cost of global structure learning. To handle the mixed continuous-categorical nature of the data and avoid distributional assumptions, all continuous variables are discretized into three equiprobable categories (low, medium, high) using the quantile method prior to running the algorithm.

Conditional dependence is evaluated via the G^2 likelihood-ratio test (equivalent to the mutual information test for discrete data):

$$G^2 = 2 \sum_{i,j,k} O_{ijk} \ln \left(\frac{O_{ijk}}{E_{ijk}} \right)$$

where i, j, k index states of the candidate variable, the target T, and the current blanket $B(T)$ respectively; O_{ijk} is the observed count and E_{ijk} the expected count under conditional independence. The algorithm proceeds in two phases:

Algorithm 1 Fast-IAMB [7]

```
Input: Target T, Variables V, Dataset D
Phase I: Forward (Speculative) Phase
Initialize B(T) ← ∅
repeat
    Find X ∈ V \ B(T) maximizing the G² statistic with T given B(T)
    if X ⫫̸ T | B(T) (via G² test) then
        B(T) ← B(T) ∪ {X}
    end if
until No variable is added
Phase II: Backward (Elimination) Phase
for each X ∈ B(T) do
    if X ⫫ T | B(T) \ {X} (via G² test) then
        B(T) ← B(T) \ {X}
    end if
end for
Return: B(T)
```

3.3 Data-Adaptive Selection via LASSO

Standard LASSO variable selection applied only to the outcome model can introduce omitted variable bias: if a variable is a weak predictor of Y but a strong driver of A, it constitutes a confounder that LASSO may drop, leaving an open backdoor path. The Post-Double-Selection procedure [2] prevents this by fitting two separate ℓ_1-penalized logistic regression models, one for Y and one for A, and taking the union of the selected variables. For a generic binary target $T \in \{Y, A\}$:

$$\hat{\boldsymbol{\beta}}^{(T)} = \arg\min_{\boldsymbol{\beta}} \left(-\frac{1}{n} \sum_{i=1}^{n} \log P(T_i \mid \mathbf{X}_i, \boldsymbol{\beta}) + \lambda_T \|\boldsymbol{\beta}\|_1 \right)$$

where λ_T is selected by 10-fold cross-validation to minimise held-out deviance. The adjustment set is $\mathbf{L} = \hat{S}_Y \cup \hat{S}_A$, where $\hat{S}_T = \{j : \hat{\beta}_j^{(T)} \neq 0\}$.

4 Estimation and Inference

Given an adjustment set $\mathbf{L}$, the ACE is estimated via logistic regression. For a binary target $Z \in \{0, 1\}$ and covariates $\mathbf{X}$, we model $\text{logit}(\mathbb{P}(Z = 1 \mid \mathbf{X})) = \boldsymbol{\beta}^\top \mathbf{X}$ and obtain $\hat{\boldsymbol{\beta}}$ by maximum likelihood. We use two complementary estimators; agreement between them strengthens confidence that the result is not an artefact of one particular model:

1. Outcome Regression (G-computation): Let $Q(\mathbf{L}_i, a; \hat{\boldsymbol{\beta}}) = \mathbb{P}(Y = 1 \mid A = a, \mathbf{L}_i; \hat{\boldsymbol{\beta}})$ be the predicted default probability. G-computation marginalises over the empirical covariate distribution:

$$\widehat{\text{ACE}}_{\text{OR}} = \frac{1}{n}\sum_{i=1}^{n}\Big(Q(\mathbf{L}_i, 1; \hat{\boldsymbol{\beta}}) - Q(\mathbf{L}_i, 0; \hat{\boldsymbol{\beta}})\Big)$$

This estimator is consistent if the outcome model is correctly specified.

2. Inverse Probability Weighting (IPW): Let $\hat{\pi}(\mathbf{L}_i) = \mathbb{P}(A = 1 \mid \mathbf{L}_i; \hat{\boldsymbol{\beta}})$ be the estimated propensity score, truncated to $[0.001, 0.999]$ to prevent numerical instability from near-zero denominators:

$$\widehat{\text{ACE}}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{A_i Y_i}{\hat{\pi}(\mathbf{L}_i)} - \frac{(1 - A_i)Y_i}{1 - \hat{\pi}(\mathbf{L}_i)}\right)$$

This estimator is consistent if the propensity score model is correctly specified. When $\mathbf{L}$ is high-dimensional, extreme propensity scores inflate the variance of the IPW estimator, directly reflecting practical positivity violations.

4.1 Uncertainty Quantification

Statistical uncertainty is quantified via a non-parametric bootstrap ($B = 500$ resamples). In each iteration, the outcome and propensity score models are refitted on the resampled data for all four adjustment sets. The adjustment sets themselves are held fixed across iterations. For the LASSO strategy, this is theoretically justified by the post-selection inference guarantees of [2], which ensure that valid inference on the treatment effect can be conducted conditional on the selected set. For the structural methods (Rank PC and Fast-IAMB), the graph is fixed because structure learning algorithms are not designed to be bootstrapped: the Rank PC algorithm is consistent for a fixed population distribution [3], but re-running it on each resample would conflate the variance of the causal estimators with the instability of graph estimation under small samples, making the resulting CIs difficult to interpret. Holding the graph fixed therefore isolates the quantity of interest, the uncertainty in the ACE given a fixed causal structure, at the cost of not propagating graph uncertainty into the intervals. Note that for the LASSO strategy, holding the set fixed may also underestimate total uncertainty given the moderate sample size ($n = 500$), as variable selection itself is a source of randomness. The 95% confidence intervals are derived via the percentile method.

5 Results and Discussion

A naive comparison of default rates yields $\hat{\Delta}_{\text{naive}} = \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0] \approx 0.132$, confirming substantial selection bias: customers who enroll in a PPP are observationally more likely to default, but this raw difference conflates self-selection with any genuine causal effect and cannot answer our first research question.

The adjustment sets identified by each strategy are detailed in Table 2, and the CPDAG recovered by Rank PC is shown in Figure 1. The payment status variables form a densely connected temporal subgraph, but only `PAY_0` has a direct edge into Y; the others influence Y solely through `PAY_0`. This Markov-shielding property is why the Rank PC adjustment set retains only `PAY_0` from the payment history block, together with `LIMIT_BAL` and `PAY_AMT2` (children of Y in the CPDAG)and `BILL_AMT3` and `PAY_AMT6` (spouses of Y), five variables in total. Fast-IAMB recovers an even sparser boundary of two variables (`PAY_0` and `PAY_AMT4`), while LASSO selects 16 variables by design agnostic to the causal graph.

Table 2: Adjustment sets selected by each strategy. The Full Model (not shown) uses all 24 covariates. Structural methods identify the Markov Boundary of Y; LASSO takes the union of variables selected for Y and A separately.

LASSO (15 variables)	**Rank PC (5 variables)**	**Fast-IAMB (2 variables)**
`LIMIT_BAL, AGE, SEX`	`LIMIT_BAL`	`PAY_0`
`PAY_0, PAY_2, PAY_3, PAY_6`	`PAY_0`	`PAY_AMT4`
`BILL_AMT6`	`PAY_AMT2`	
`PAY_AMT1` to `PAY_AMT6`	`BILL_AMT3`	
`EDU_University`	`PAY_AMT6`	

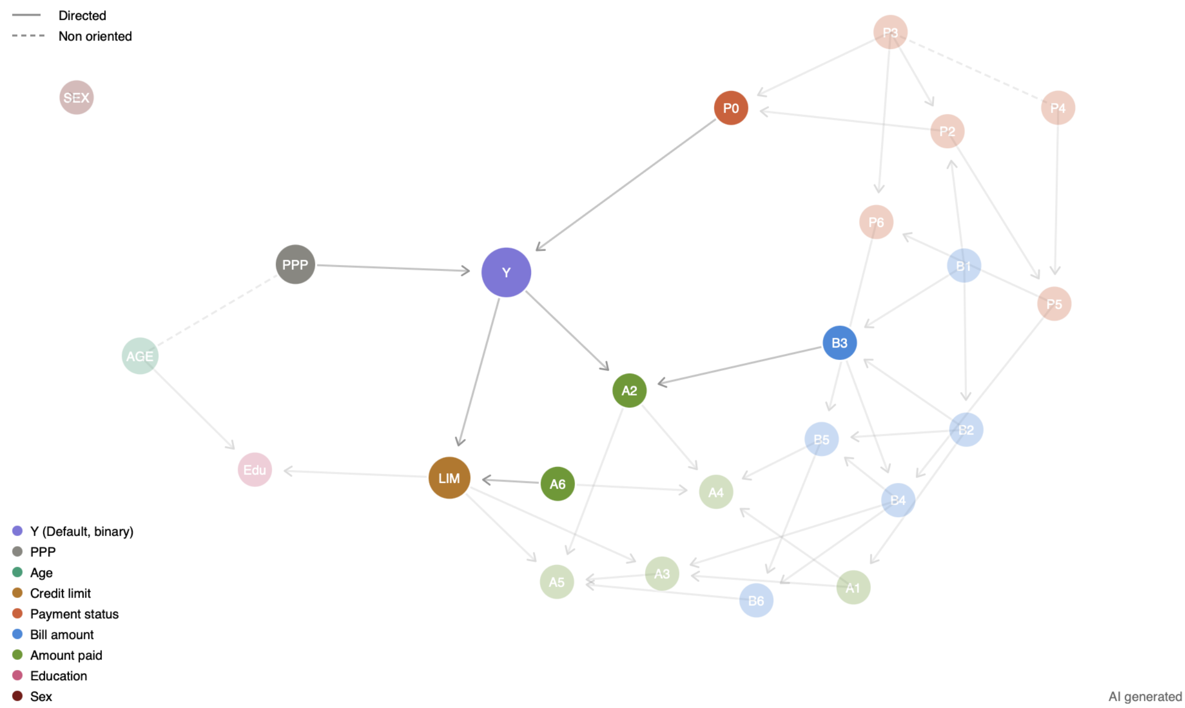

Figure 1: CPDAG recovered by the Rank PC algorithm ($\alpha = 0.05$, nonparanormal transformation). Solid arrows are directed edges; dashed lines are undirected edges whose orientation could not be determined from the data. Nodes are colour-coded by variable type. The Markov blanket of Y (parents `PAY_0`, `PPP`; children `LIMIT_BAL`, `PAY_AMT2`; spouses `BILL_AMT3`, `PAY_AMT6`) is visually highlighted.

Point estimates and bootstrap 95% CIs for all eight estimator–adjustment combinations are in Table 3 and Figure 2.

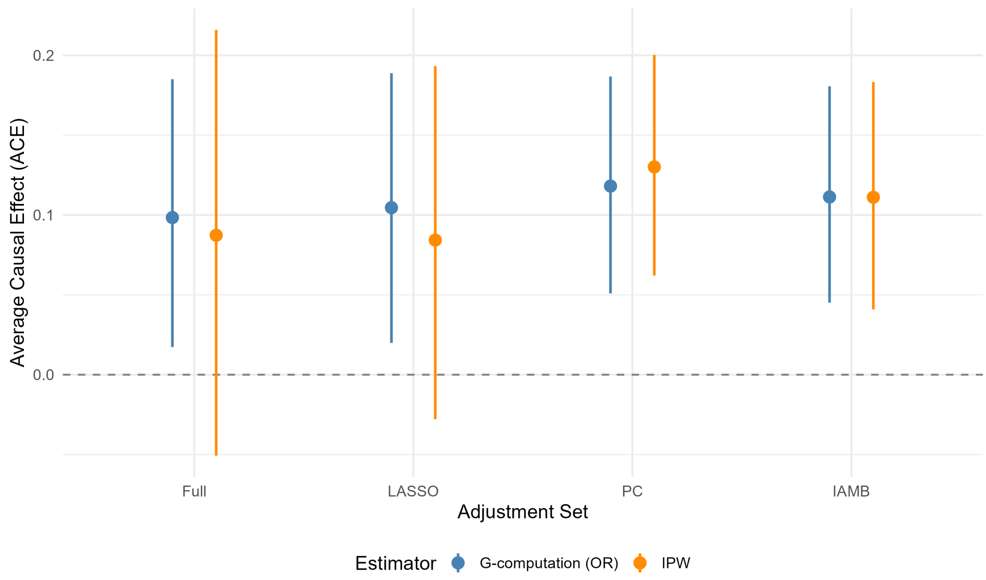

Figure 2: Point estimates (dots) and 95% bootstrap percentile confidence intervals (bars) for the ACE of PPP enrollment on default probability, across four adjustment sets and two estimators. The dashed horizontal line marks ACE = 0. OR = Outcome Regression (G-computation); IPW = Inverse Probability Weighting. $B = 500$ resamples.

Table 3: Bootstrap results ($B = 500$ resamples, percentile method). Width = upper bound − lower bound of the 95% CI. Intervals crossing zero are shown in bold.

Estimator	**Adjustment**	**ACE**	**95 % CI**	**Width**
OR	Full	0.098	[0.017, 0.185]	0.168
	LASSO	0.105	[0.020, 0.189]	0.169
	Rank PC	0.118	[0.051, 0.187]	0.136
	Fast-IAMB	0.111	[0.045, 0.181]	0.136
IPW	Full	0.087	[**−0.051**, **0.216**]	0.267
	LASSO	0.084	[**−0.028**, **0.193**]	0.221
	Rank PC	0.130	[0.062, 0.200]	0.138
	Fast-IAMB	0.111	[0.041, 0.183]	0.142

5.1 Positivity and Variance Inflation

The results expose a fundamental tension between predictive accuracy and causal validity. Under IPW, the Full and LASSO adjustment sets yield CIs that cross zero (Table 3), rendering the effect statistically indistinguishable from null despite positive point estimates. The mechanism is a practical positivity violation: conditioning on 24 (Full) or 16 (LASSO) variables allows the propensity model to predict treatment assignment with high accuracy for some individuals, driving propensity scores close to 0 or 1 and assigning them extreme inverse weights. This is not a failure of IPW as a method, but a signal that the adjustment set is too rich relative to $n = 500$. The Rank PC and Fast-IAMB sets (5 and 2 variables respectively) are sufficient to block the relevant backdoor paths while preserving well-behaved propensity scores, yielding CI widths of 0.136–0.142 under both estimators, roughly half those of the Full model. This directly answers our second research question: causal discovery does not merely provide a theoretically principled adjustment set, it also yields strictly more efficient estimates by preserving the positivity assumption.

5.2 Causal Interpretation

Across all correctly specified structural models, the ACE is consistently positive and statistically significant. Adjusting for the estimated confounding bias of approximately 3.4 p.p. ($\hat{\Delta}_{\text{naive}} - \widehat{\text{ACE}}_{\text{OR, Full}} \approx 0.034$), the pure causal effect of PPP enrollment on default probability ranges from approximately +10 to +13 percentage points, directly answering our first research question: PPPs increase, rather than decrease, the probability of default.

These findings are consistent with a moral hazard mechanism [1]: when customers are shielded from the immediate financial cost of missing a payment, the incentive to manage liquidity carefully is weakened, ultimately increasing default probability. Stiglitz and Weiss [6] formalised this logic in the credit market, showing that reducing the effective cost of non-repayment shifts borrowers toward riskier strategies. The present results are consistent with this interpretation, though the simulated nature of the dataset precludes definitive causal attribution to this specific behavioral channel.

6 Conclusion

This study provides causal evidence that Payment Protection Plans increase the probability of credit default. After controlling for confounding through structural causal discovery, the estimated ACE ranges from +11 to +13 p.p., with CIs excluding zero across all correctly specified adjustment sets.

Furthermore, causal discovery methods (Rank PC, Fast-IAMB) recover parsimonious adjustment sets that satisfy the positivity assumption and yield confidence intervals roughly half as wide as those from high-dimensional predictive adjustment. In observational causal inference, parsimony in covariate selection is a requirement for valid inference.

6.1 Limitations

Four limitations deserve mention. First, the dataset contains only $n = 500$ observations, limiting statistical power and making high-dimensional adjustment fragile. Second, causal identification rests on conditional exchangeability (no unmeasured confounders); variables such as individual risk tolerance are unobserved and, if correlated with both PPP enrollment and default propensity, would bias the ACE upward, reinforcing but potentially overstating the positive effect found here. Third, the Rank PC graph is sensitive to α and the specific sample; graph uncertainty is not propagated into the bootstrap CIs. Fourth, both estimators assume a logistic outcome model (linearity in the log-odds), which was not formally validated; misspecification could bias the OR estimates while leaving IPW estimates consistent, so the convergence between OR and IPW across structural adjustment sets provides partial reassurance.

6.2 Practical Implications

These findings suggest that PPP enrollment should be treated as a risk indicator rather than a mitigant. Financial institutions may benefit from segmenting customers before enrollment, identifying profiles for which the behavioral response to payment deferral is more pronounced, and pricing protection plans to reflect the increased default risk they may induce.

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